Machine Learning: k-Means and PCA for Regime Detection

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1. Abstract

In this project, we apply k-means clustering and PCA to solve the problem of market regime detection in multi-dimensional financial time-series. We compare these techniques against other regime detection methods: the Hidden Markov Model with Gaussian Mixtures (GMMHMM), sliced-Wasserstein k-means clustering, and a simple peak-to-trough-based regime labelling method. We evaluate the quality of each detection method using a trading strategy that goes long when the label indicates a bullish regime and flat otherwise, comparing cumulative returns across methods and regions (America, Europe, Asia).

2. Data Processing and Feature Engineering

We track 9 major stock indices across 3 regions to develop our analysis.

Table 1: Index universe by region.

Region	Index	Ticker
America	S&P 500 / Dow Jones / Nasdaq	`^GSPC`, `^DJI`, `^IXIC`
Europe	FTSE 100 / CAC 40 / DAX	`^FTSE`, `^FCHI`, `^GDAXI`
Asia	KOSPI / Nikkei 225 / Hang Seng	`^KS11`, `^N225`, `^HSI`

2.1. Raw Data Structure

Each index is downloaded from Yahoo Finance with daily OHLCV columns. After merging, the full DataFrame uses a MultiIndex on columns of the form `(index_name, OHLCV_field)`.

2.2. Feature Engineering

From the closing prices we compute the following features for each index:

Log-returns. Over multiple horizons (1, 5, and 21 trading days):

$$r_{t,p} = \ln\left(\frac{P_t}{P_{t-p}}\right). \tag{1}$$

Rolling annualised volatility. With windows of 21 and 63 days:

$$\sigma_{t,w} = \mathrm{std}(r_{t-w:t}) \times \sqrt{252}\,. \tag{2}$$

Rolling correlation. For each pair of indices X and Y at time t with window w:

$$\rho_{X,Y,t} = \frac{\sum_{i=0}^{w-1}(r_{X,t-i} - \bar{r}_X)(r_{Y,t-i} - \bar{r}_Y)}{\sqrt{\sum_{i=0}^{w-1}(r_{X,t-i} - \bar{r}_X)^2 \ \sum_{i=0}^{w-1}(r_{Y,t-i} - \bar{r}_Y)^2}} \tag{3}$$

Momentum Z-score. Measures mean-reversion by normalising distance to the moving average:

$$Z_{t,w} = \frac{P_t - \mathrm{MA}_{t,w}}{\sigma_{t,w}} \tag{4}$$

RSI. The Relative Strength Index captures overbought and oversold conditions:

$$\mathrm{RSI}_t = 100 - \frac{100}{1 + \frac{\text{Average Gain}_t}{\text{Average Loss}_t}} \tag{5}$$

where Average Gain_t and Average Loss_t are computed over 14 days.

Building the feature set. We merge all features (log-returns, rolling volatility, correlation, z-score, RSI) into a single DataFrame. Per index: 3 log-returns + 2 volatilities + 1 z-score + 1 RSI = 7 features; across 3 indices this gives 21 features, plus 3 pairwise correlations, yielding a raw feature matrix of 24 columns. The number of rows equals the trading days minus a 63-day warm-up period. We standardise features and apply PCA retaining 95% of variance.

Table 2: Feature set dimensions after PCA (95% variance retained).

Region	Raw dimensions	After PCA
America	$(3460, 24)$	$(3460, 7)$
Europe	$(3523, 24)$	$(3523, 10)$
Asia	$(3561, 24)$	$(3561, 14)$

2.3. Pipeline Summary

Yahoo Finance API → Raw OHLCV → Merged DataFrame → Close prices

↓

Feature Engineering (returns, volatility, correlation, z-score, RSI)

↓

Standardisation → **PCA** → **K-Means Clustering**

Figure 1: Data pipeline from raw download to regime labels.

3. Mathematical Foundations

3.1. K-Means Clustering

K-means partitions data into K clusters by minimising the within-cluster sum of squares (WCSS):

$$\mathrm{WCSS} = \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|^2 \tag{6}$$

where C_k is the set of points in cluster k and μ_k is its centroid. In high-dimensional settings, the curse of dimensionality can limit performance; we therefore reduce dimensionality via PCA before clustering.

3.2. Principal Component Analysis (PCA)

PCA finds directions of maximum variance via the singular value decomposition of the centred data matrix:

$$\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top \tag{7}$$

where $\mathbf{U}$ contains left singular vectors, $\mathbf{\Sigma}$ is a diagonal matrix of singular values, and $\mathbf{V}$ contains the principal components. We retain the top d^* components capturing $\geq 95\%$ of variance.

4. Proposed Algorithm

Algorithm 1 K-Means Regime Detection Pipeline

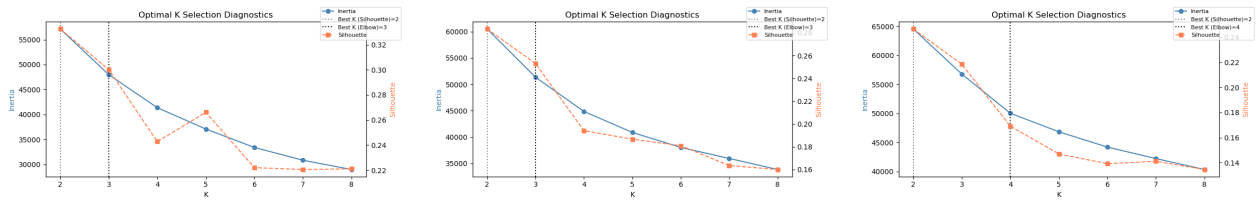
Require: Closing prices $\mathbf{P} \in \mathbb{R}^{T \times N}$, variance threshold α, range $[K_{\min}, K_{\max}]$
Ensure: Regime labels $\ell_t \in \{0, \ldots, K^* - 1\}$, ordered by mean return

— **Stage 1: Feature Engineering** —
for each index $n = 1, \ldots, N$ **do**
  Compute log-returns $r_{t,p}^{(n)}$ for $p \in \{1, 5, 21\}$
  Compute rolling volatility $\sigma_{t,w}^{(n)}$ for $w \in \{21, 63\}$
  Compute momentum z-score $Z_t^{(n)}$ and $\mathrm{RSI}_t^{(n)}$
end for
for each pair (i, j) with $i < j$ **do**
  Compute $\rho_t^{(i,j)} \leftarrow \mathrm{RollingCorr}_{63}(r^{(i)}, r^{(j)})$
end for
Concatenate into $\mathbf{F} \in \mathbb{R}^{T' \times D}$; drop warm-up rows
— **Stage 2: Standardisation & PCA** —
Z-score normalise each feature; compute SVD of $\tilde{\mathbf{F}}$
Select $d^* = \min\{d : \sum_{i=1}^{d} \sigma_i^2 / \sum_{i=1}^{D} \sigma_i^2 \geq \alpha\}$; project $\mathbf{X} \leftarrow \tilde{\mathbf{F}}\, \mathbf{V}_{:,1:d^*}$
— **Stage 3: Optimal K & Clustering** —
for $K = K_{\min}, \ldots, K_{\max}$ **do**
  Run K-Means; record inertia $\mathcal{I}(K)$ and silhouette $\mathcal{S}(K)$
end for
$K^* \leftarrow \arg\max_K \mathcal{S}(K)$; run final K-Means with K^*
— **Stage 4: Regime Ordering** —
Compute mean basket return $\bar{r}_k$ per cluster; relabel $0 =$ bearish, $K^* - 1 =$ bullish
return $\boldsymbol{\ell} = (\ell_1, \ldots, \ell_{T'})$

5. Results

5.1. Optimal K Selection

We applied k-means for $K \in \{2, \ldots, 10\}$ on the PCA-reduced feature space and computed inertia and silhouette scores. As shown in Figure 2, the silhouette score is maximised at $K = 2$ across all regions, suggesting two dominant regimes over 2006–2020.

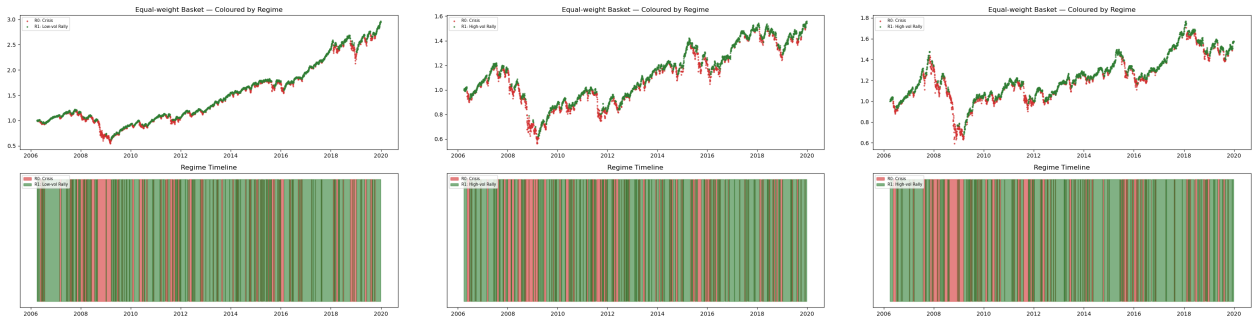


(a) American indices. **(b)** European indices. **(c)** Asian indices.

Figure 2: Inertia and silhouette scores for different K values across regional indices.

5.2. Regime Detection Results

Figure 3 shows the detected regimes overlaid on an equally weighted basket of indices for each region. The algorithm captures major market downturns (2008 crisis, 2011 European debt crisis, 2015–16 volatility) with high precision.

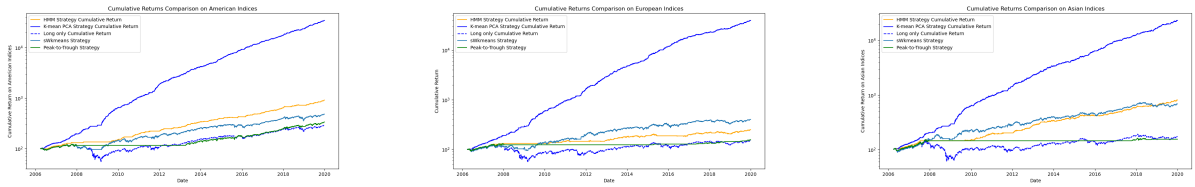


(a) American indices. **(b)** European indices. **(c)** Asian indices.

Figure 3: Regime detection on equally weighted baskets for each region.

5.3. Trading Strategy Comparison

To evaluate regime quality, we implement a long/flat strategy: go long in bullish regimes, hold cash otherwise. We benchmark PCA K-Means against GMMHMM, sliced-Wasserstein k-means, peak-to-trough labelling (20% drawdown threshold), and a long-only baseline. All strategies start with $100 over March 2006 – January 2020.



(a) American indices. **(b)** European indices. **(c)** Asian indices.

Figure 4: Cumulative returns of trading strategies based on different regime detection methods.

The PCA K-Means strategy significantly outperforms all alternatives across every region, as summarised in Table 3.

Table 3: Backtest performance summary across regional indices.

Region	Metric	PCA K-Means	HMM	sWK-Means	Peak-to-Trough	Long Only
American	Cumulative Return (%)	34,808.39	915.48	483.31	337.18	293.05
	Annualised Return (%)	53	16	12	9	8
	Annualised Volatility (%)	10	6	15	10	19
	Sharpe Ratio	4.37	2.74	0.85	0.95	0.51
European	Cumulative Return (%)	39,532.73	247.17	401.41	155.08	149.26
	Annualised Return (%)	53	7	10	3	3
	Annualised Volatility (%)	11	7	15	6	20
	Sharpe Ratio	3.91	0.94	0.72	0.57	0.24
Asian	Cumulative Return (%)	23,225.90	810.79	687.00	155.36	173.54
	Annualised Return (%)	47	15	15	3	4
	Annualised Volatility (%)	10	8	14	6	19
	Sharpe Ratio	3.76	1.90	1.04	0.54	0.30

Looking at the performance metrics, the PCA K-Means strategy achieves annualised returns of 47–53% with Sharpe ratios above 3.7 across all regions, while alternatives lag significantly behind. The long-only benchmark delivers much lower returns and Sharpe ratios. We can therefore see that PCA K-means capture market regimes in a highly granuar way, as opposed to other methods which are either too noisy (sW k-means) or too coarse (peak-to-trough). The HMM performs decently but still falls short of the PCA K-Means approach, likely due to its parametric assumptions and sensitivity to regime duration. However, while we present performance metrics for trading strategies, those results have to be taken as a descriptive accuracy rather than a predictive accuracy, as for all regime detection methods, we have high look-ahead bias when labelling regimes. This is because we

are using the entire dataset to detect regimes, and then backtesting a strategy on the same dataset. Therefore, while the PCA K-Means method shows superior performance in this backtest, further work would be needed to evaluate its predictive power in an out-of-sample context.

5.4. Hyperparameter tuning

6. Conclusion

We have presented a PCA-enhanced k-means clustering pipeline for market regime detection across American, European, and Asian equity indices. The method consistently identifies two dominant regimes (bullish and bearish) that align well with known periods of market stress. When used to drive a simple long/flat trading strategy, the PCA K-Means approach delivers substantially higher cumulative returns and Sharpe ratios than alternative methods (GMMHMM, sliced-Wasserstein k-means, peak-to-trough labelling) and the long-only benchmark across all three regions. These results suggest that combining dimensionality reduction with unsupervised clustering provides a robust framework for capturing market dynamics. Additionally ...

Future work could explore out-of-sample regime prediction, rolling re-estimation windows, and the inclusion of additional asset classes to further validate the approach.

7. Team Work and Contributions

7.1. Individual Contribution

- Zhidian Jing:
- Siqi Zhang:
- Stefaan Fernando:
- Daniel Greenstein:
- Oscar Peyron: Implemented the Data Processing and Feature Engineering Pipeline. Implemented the PCA and K-means clustering framework.

7.2. Reflection on Team Work